

PAPER CODE: BEM-C102
ODD SEMESTER EXAMINATION
(DECEMBER-2025)
CLASS: B.TECH. SEMESTER: I SEM
PAPER TITLE: ENGINEERING MATHEMATICS-I

Time: 3 hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 x 6 = 30 Marks)

Question-1: Find the n^{th} derivative of $\cos x \cdot \cos 2x \cdot \cos 3x$.

Question-2: Find the radius of curvature to the cycloid $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$ at the point θ . What will it be at $\theta = \pi$.

Question-3: If $u = \tan^{-1} \left(\frac{x^2 + y^2}{x - y} \right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$.

Question-4: Verify whether $u = \sin^{-1} x + \sin^{-1} y$ and $v = x\sqrt{1 - y^2} + y\sqrt{1 - x^2}$ are functionally dependent, and if so find the relation between them.

Question-5: Evaluate by changing the order of integration $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$.

Question-6: Evaluate $\int_0^1 x^{n-1} \left[\log \left(\frac{1}{x} \right) \right]^{m-1} dx$.

Question-7: Use Green's Theorem to evaluate $\int_c [(x^2 + xy)dx + (x^2 + y^2)dy]$ where c is square formed by the lines $x = \pm 1$, $y = \pm 1$.

Question-8: State and prove physical significance of CURL

Question-9: Show that the matrix B^0AB is Hermitian or skew Hermitian according as A is Hermitian or skew Hermitian.

Question-10: Find the rank of the matrix by reducing it to normal form $\begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 4 & 8 & 13 & 12 \end{bmatrix}$.

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any **four** questions in detail. Each question carries 10 marks.
(4 x 10 = 40 Marks)

Question-1: If $\cos^{-1}\left(\frac{y}{b}\right) = \log\left(\frac{x}{n}\right)^n$, prove that

$$x^2 y_{n+2} + (2n+1)xy_{n+1} + 2n^2 y_n = 0.$$

Question-2: A function $f(x,y)$ is rewritten in terms of new variables $u = e^x \cos y$, $v = e^x \sin y$, prove

$$\text{that } \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = (u^2 + v^2) \left(\frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2} \right).$$

Question-3: Use the method of Lagrange's multipliers to find volume of largest

rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

Question-4: Find the area between the parabola $y^2 = 4ax$ and $x^2 = 4ay$.

Question-5: Evaluate $\iiint_S \frac{dx dy dz}{(x+y+z+1)^3}$, where S is bounded by the coordinate planes and the plane $x + y + z = 1$.

Question-6: Verify Stoke's theorem for $\vec{F} = (x+y)\mathbf{i} + (2x-z)\mathbf{j} + (y+z)\mathbf{k}$ for the surface of a triangular lamina with vertices $(2,0,0)$, $(0,3,0)$, $(0,0,6)$.

Question-7: Test for consistency and solve $x + y + z = 6$, $x + 2y + 3z = 14$, $x + 4y + 7z = 30$

Question-8: Find the directional derivative of the function $F = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of the line PQ where Q is the point $(5, 0, 4)$.